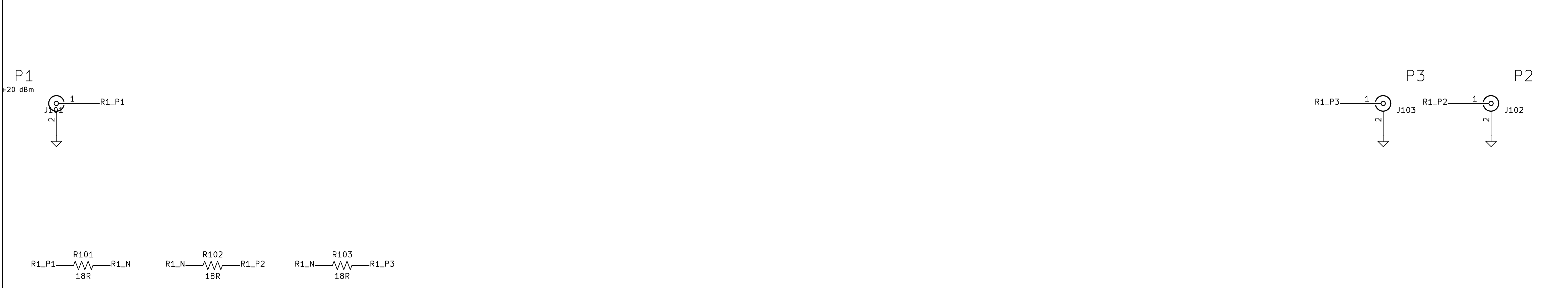
R1 splitter, 3 x 18 ohm, 6 dB

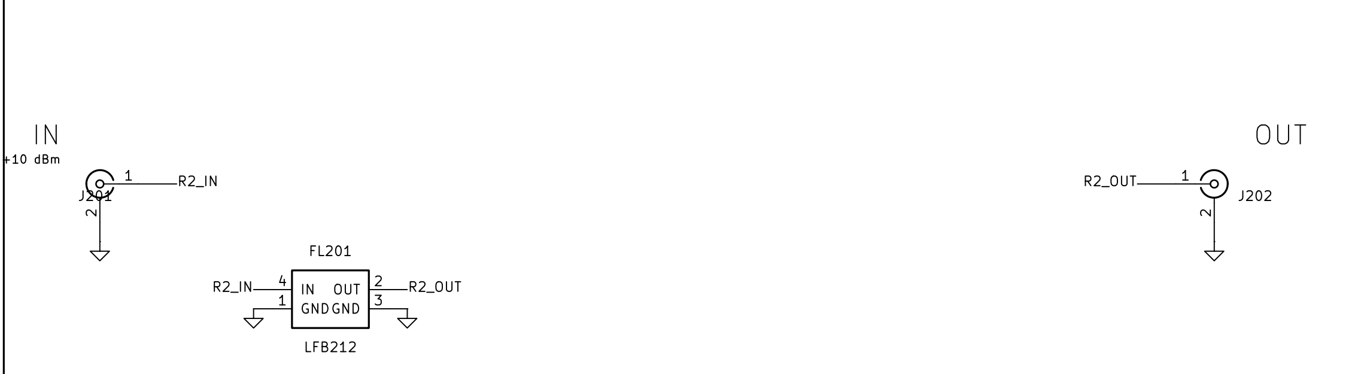
cells (0,0), (1,0)


P1
+20 dBm
J101
R1_P1
P3
R1_P3
J103
P2
R1_P2
J102
R101
18R
R102
18R
R103
18R
R1_N

EXPECT: 6.2 dB from any port to either other port, every port within 2 ohm of 50, DC to 6 GHz (the resistors' parasitics set the top)
Three 18 ohm resistors to one node: the textbook resistive divider, 6 dB per path and no isolation between the two outputs beyond those 6 dB (a Wilkinson would give 3 dB and 20 dB of isolation, at one band). Feed R9's REF OUT to both synthesizers' REF IN, one synthesizer's OUT A to both mixers, or two tones into one gain block.

R2 2.45 GHz filter

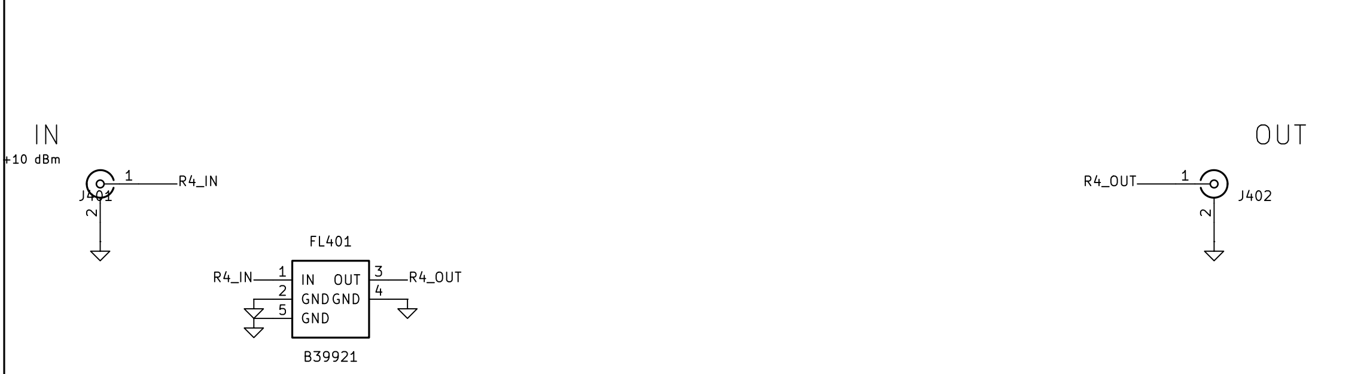
cell (2,0)


IN
+10 dBm
J201
R2_IN
OUT
R2_OUT
J202
FL201
LFB212

EXPECT: 1.7 dB of loss from 2.4 to 2.5 GHz, 30 dB or more below 2.2 GHz and above 2.7 GHz (the datasheet); 50 ohm in and out
Murata LFB212G45SG8A127, a 2012-size LTCC band-pass. The line to each jack is 50 ohm coplanar waveguide (0.29 mm wide, 0.25 to the ground pour) with a via fence; the jack's centre pin has no plane under it (a 2.2 mm void).
Sweep it with R10 or R13 into a power meter, or with a DAC harmonic comb and the ADC.

R4 915 MHz filter

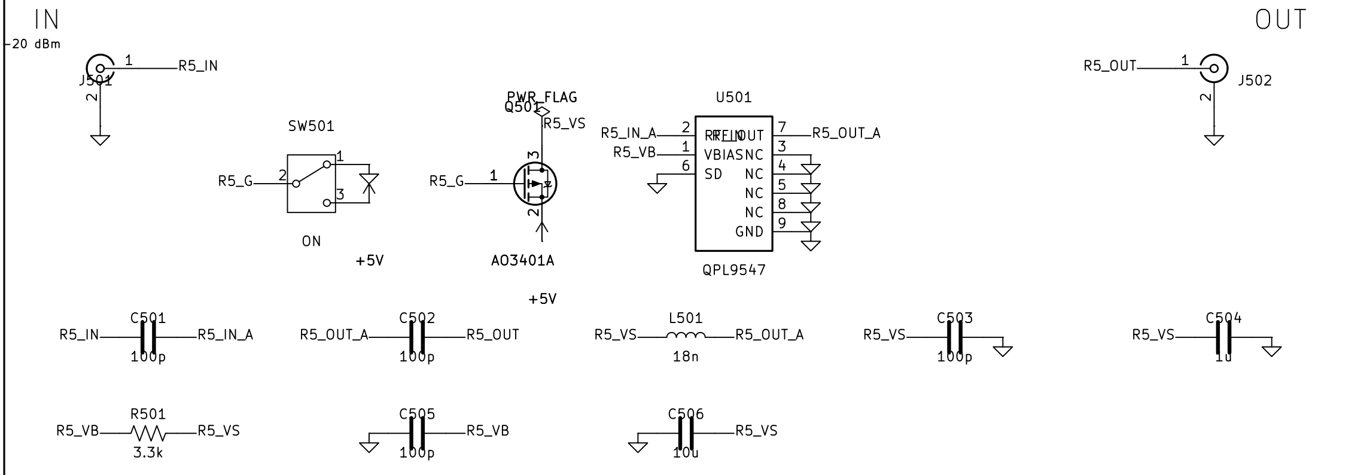
cell (3,0)


IN
+10 dBm
J401
R4_IN
OUT
R4_OUT
J402
FL401
B39921

EXPECT: 1.1 dB of loss from 902 to 928 MHz, 40 dB or more 30 MHz outside the band (the datasheet); 50 ohm in and out
RF360 B39921B2672P810: a surface-acoustic-wave filter for the 915 MHz ISM band, 26 MHz wide, in a 1411 package; its skirts are much steeper than the LTCC's. Lines, fences and launch voids as on R2.

R5 LNA, QPL9547

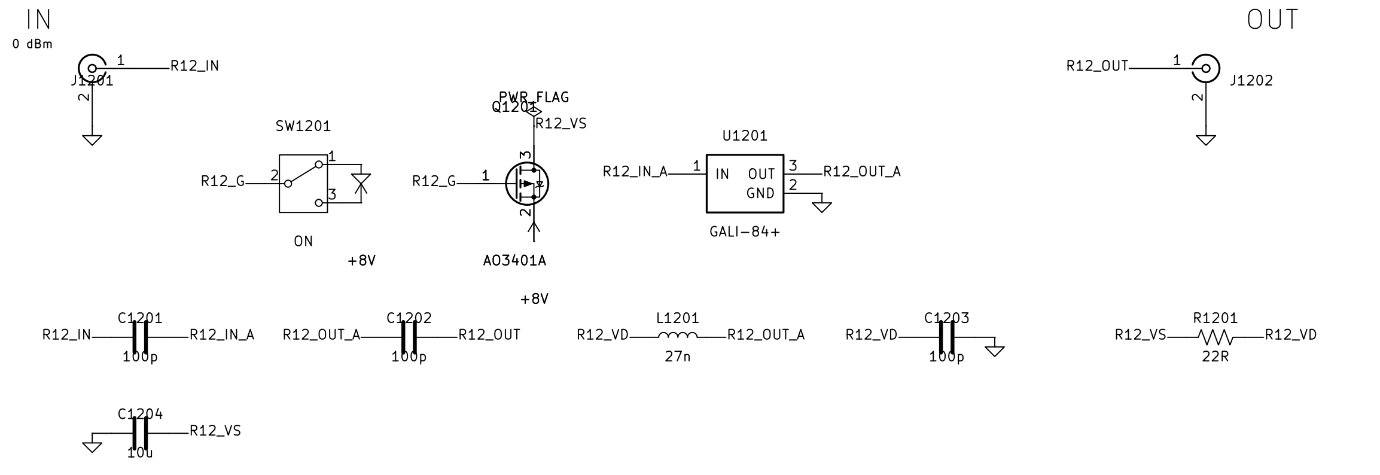
cell (0,1)


IN
-20 dBm
J501
R5_IN
SW501
ON
Q501
AO3401A
PWR_FLAG
+5V
U501
QPL9547
OUT
R5_OUT
J502
L501
18n
R501
3.3k
R5_VB
R5_VS
100p
10u

EXPECT: 19.5 dB of gain and a 0.3–0.5 dB noise figure from 0.5 to 3 GHz (datasheet: 0.3 dB at 1.9 GHz, OIP3 +39 dBm); 65 mA from 5 V
Qorvo QPL9547: a GaAs pHEMT LNA, internally matched to 50 ohm, 0.1–6 GHz. It sets the noise figure of every receiver here, so put it first (before the filter) unless a strong neighbour overloads it; +39 dBm OIP3 makes that rare.
The 18 nH wire-wound choke (Qorvo's value) feeds 5 V into the output pin; 3.3 k from 5 V to Vbias sets 65 mA.
Never cable a transmitter into it without 30 dB of attenuation (22 dBm absolute maximum at the input).

R12 gain block, GALI–84+

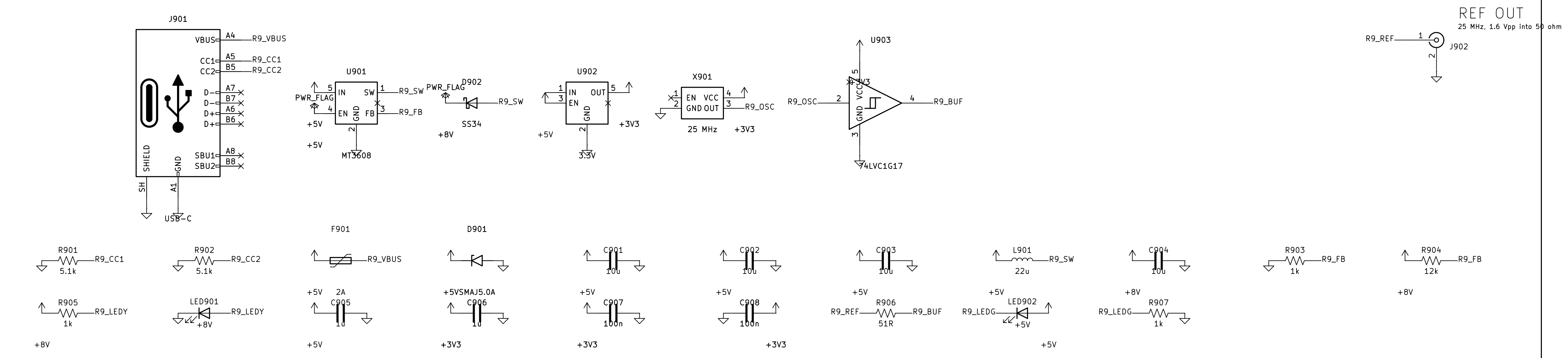
cell (1,1)


IN
0 dBm
J1201
R12_IN
SW1201
ON
Q1201
AO3401A
PWR_FLAG
+8V
U1201
GALI-84+
OUT
R12_OUT
J1202
L1201
27n
R1201
22R
R12_VS
R12_VD
100p
10u

EXPECT: 20 dB of gain from 50 MHz to 2.5 GHz (datasheet: 21.9 dB at 1 GHz, 1 dB compression at +20 dBm out); 97 mA from 7.8 V
Mini–Circuits GALI–84+: a Darlington MMIC that sets its own current; the 22 ohm from the 7.8 V rail drops to its 5.8 V device voltage (0.2 W in the 2512). The 27 nH choke (415 ohm at 2.45 GHz, 155 at 915 MHz) feeds the output pin; 100 pF blocks keep the DC off the jacks. Keep the input under 0 dBm for linear gain. The slide switch and P–FET gate the tile's 7.8 V as on AMOD–A.

R9 power, USB–C 5 V in: +5, +7.8 V; 25 MHz reference

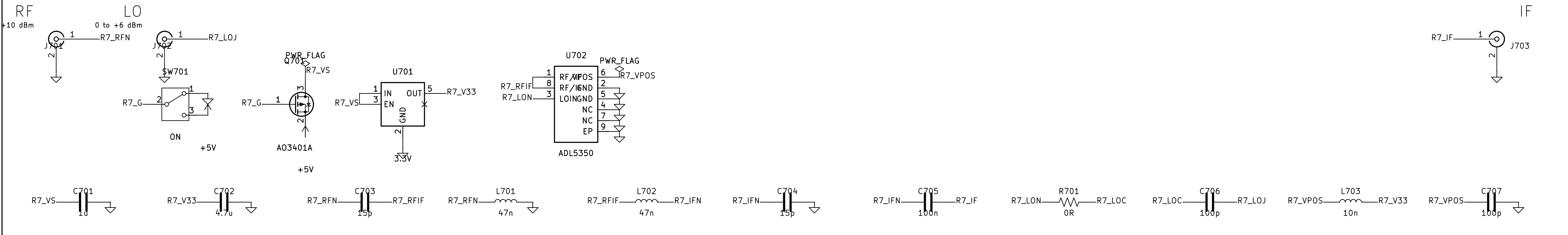
cells (2,1), (3,1)


J901
USB-C
R9_VBUS
R9_CC1
R9_CC2
U901
MT3608
R9_SW
R9_FB
D902
SS34
U902
X901
25 MHz
U903
74LVC1G17
R9_OSC
R9_BUF
REF OUT
25 MHz, 1.6 Vpp into 50 ohm
J902
R9_REF
R901
5.1k
R902
5.1k
F901
2A
D901
SMAJ5.0A
R905
1k
LED901
R9_LEDY
L901
22u
R903
1k
R904
12k
R906
51R
LED902
R907
1k
+5V
+8V
+3V3

EXPECT: +5 V from the USB–C (up to 1.5 A), +7.8 V at 250 mA for the gain blocks (yellow LED), a 25 MHz square wave on REF OUT
USB–C power only: 5.1 k on each CC asks any source for 5 V; a 2 A PTC, a 5 V TVS and 20 uF. The MT3608 boosts to 7.8 V (0.6 V x 13 from 12 k / 1 k) for the GALI–84+ tile, R12 (100 mA); the +8V rail is a strip under row 1 that their headers reach with a via. The TPS7A2033 makes 3.3 V for the TC32H4 TCXO (+–2 ppm) and the 74LVC1G17 that drives REF OUT (51 ohm: 1.6 Vpp into 50 ohm, 3.3 Vpp into a synthesizer's unterminated REF IN; R1 feeds both synthesizers).
The TCXO's pad 1 (enable or no connection on the drawing) floats, as CostasLink's review settled.

R7 TX mixer, ADL5350

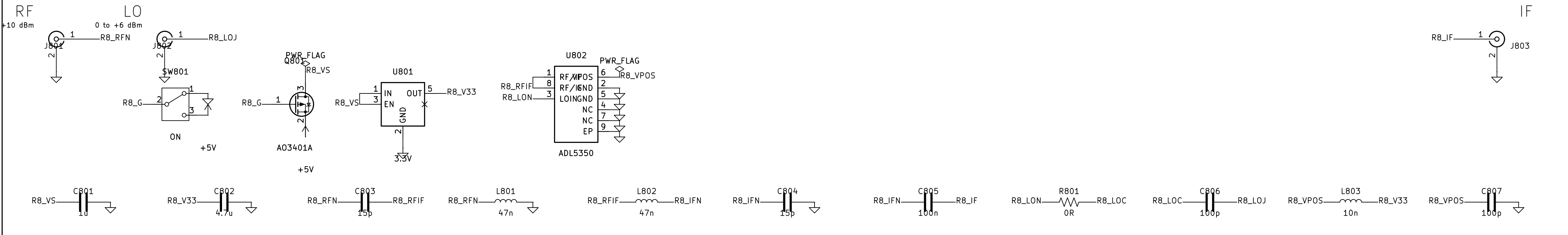
cells (0,2), (1,2)


RF
+10 dBm
J701
R7_RFN
LO
0 to +6 dBm
J702
R7_LOJ
SW701
ON
Q701
AO3401A
PWR_FLAG
+5V
U701
U702
ADL5350
IF
R7_IF
J703
L701
47n
L702
47n
R701
0R
L703
10n
15p
100n
100p
1u
4.7u

EXPECT: RF x LO to IF (or IF x LO to RF) with 7–9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1–30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band–tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3–8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up–converts a DAC tone (IF in, RF out) or down–converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with R2 or R4.

R8 RX mixer, ADL5350

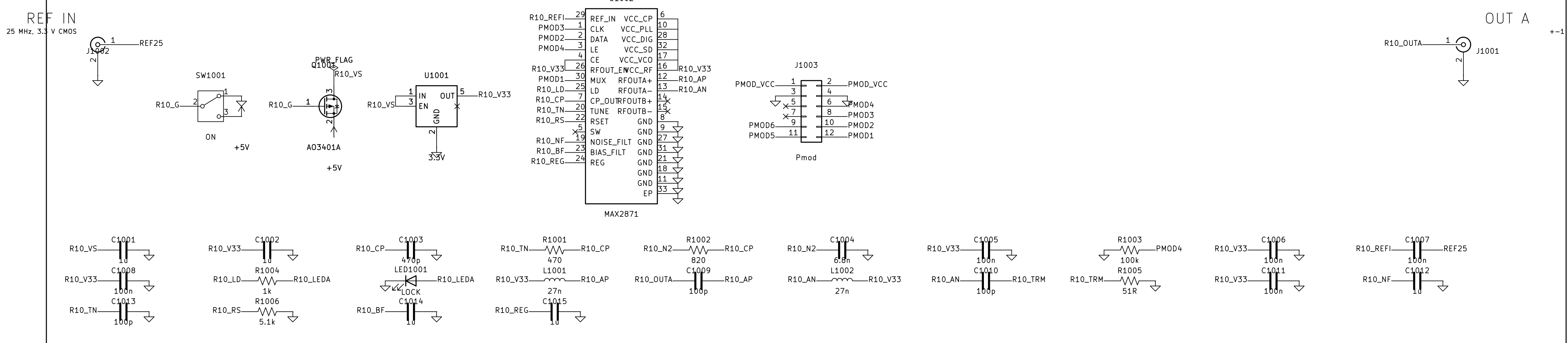
cells (2,2), (3,2)


RF
+10 dBm
J801
R8_RFN
LO
0 to +6 dBm
J802
R8_LOJ
SW801
ON
Q801
AO3401A
PWR_FLAG
+5V
U801
U802
ADL5350
IF
R8_IF
J803
L801
47n
L802
47n
R801
0R
L803
10n
15p
100n
100p
1u
4.7u

EXPECT: RF x LO to IF (or IF x LO to RF) with 7–9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1–30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band–tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3–8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up–converts a DAC tone (IF in, RF out) or down–converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with R2 or R4.

R10 synthesizer, MAX2871

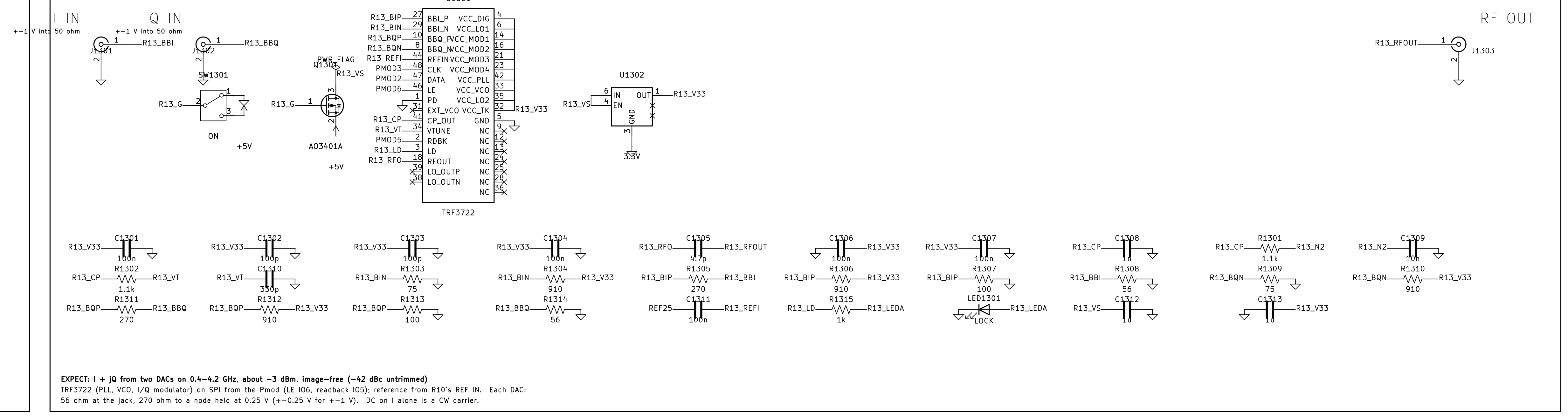
cells (0,3), (1,3)


REF IN
25 MHz, 3.3 V CMOS
J1001
REF25
SW1001
ON
Q1001
AO3401A
PWR_FLAG
+5V
U1001
U1002
MAX2871
J1003
Pmod
OUT A
R10_OUTA
J1001
C1001
1u
C1002
1u
C1003
470p
R1001
470
C1008
100n
R1004
1k
LED1001
LOCK
L1001
27n
C1013
100p
R1006
5.1k
C1014
1u
C1015
1u
R10_VS
R10_V33
R10_CP
R10_TN
R10_LD
R10_LEDA
R10_RS
R10_BF
R10_REG

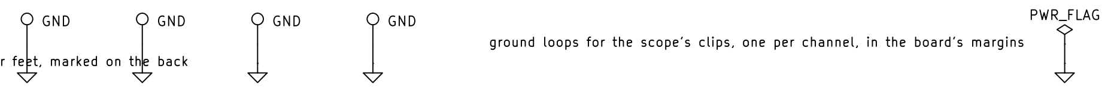
EXPECT: any frequency from 23.5 MHz to 6 GHz on OUT A (–4 to +5 dBm in steps), locked to the 25 MHz on REF IN; the red LED says locked
MAX2871 as CostasLink built it: fractional–N PLL with four VCOs, programmed over SPI from the Pmod (SCLK, SDATA, LE; MUXOUT back). The loop filter is third order, 80–120 kHz with 61 degrees of margin for both 915 MHz (VCO / 4) and 2.4 GHz (VCO / 2) at ICP 2.56 mA (RSET 5.1 k, CP = 7). RFOUTA+ feeds the jack through 27 nH to 3.3 V and 100 pF; RFOUTA– is terminated in 51 ohm the same way; RFOUTB is open and must stay disabled. CE and RFOUT_EN are tied high, LE has 100 k down. REF IN is unterminated: one REF OUT drives both synthesizers through R1.

R13 I/Q transmitter, TRF3722

cells (2,3), (3,3)


I IN
+–1 V into 50 ohm
J1301
R13_BBI
Q IN
J1302
R13_BBQ
SW1301
ON
Q1301
AO3401A
PWR_FLAG
+5V
U1301
TRF3722
U1302
RF OUT
R13_RFOUT
J1303
R1302
1.1k
C1310
330p
R1303
75
R1304
910
R1311
270
R1312
910
R1313
100
R1314
56
LED1301
LOCK
R13_CP
R13_VT
R13_BIN
R13_BQP
R13_BBQ
R13_V33
REF25

EXPECT: I + jQ from two DACs on 0.4–4.2 GHz, about –3 dBm, image–free (–42 dBc untrimmed)
TRF3722 (PLL, VCO, I/Q modulator) on SPI from the Pmod (LE IO6, readback IO5); reference from R10's REF IN. Each DAC: 56 ohm at the jack, 270 ohm to a node held at 0.25 V (+–0.25 V for +–1 V). DC on I alone is a CW carrier.


GND
GND
GND
GND
PWR_FLAG

no mounting holes: four stick–on rubber feet, marked on the back

ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/rf_mod.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a quadrature transmitter, a synthesizer, two mixers, an LNA and a gain block, two filters, a splitter, p
Board: Jason Gallicchio and Claude | open hardware, CERN–OHL–P–2.0 | github.com/gallicchio/
Sheet: /
File: rf_mod.kicad_sch
Title: RF–MOD: 915 MHz, 2.4 GHz and a stretch to 5.8 GHz
Size: A1
Date: 2026–10–25
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/1